



# Parktown Boys' High School

Geography Department

Grade 12 Control Test

Examiner: Mr A Meintjes

Time: 60 Minutes

Moderator: Mr D Grace

Total: 60 Marks

Date: 5 February

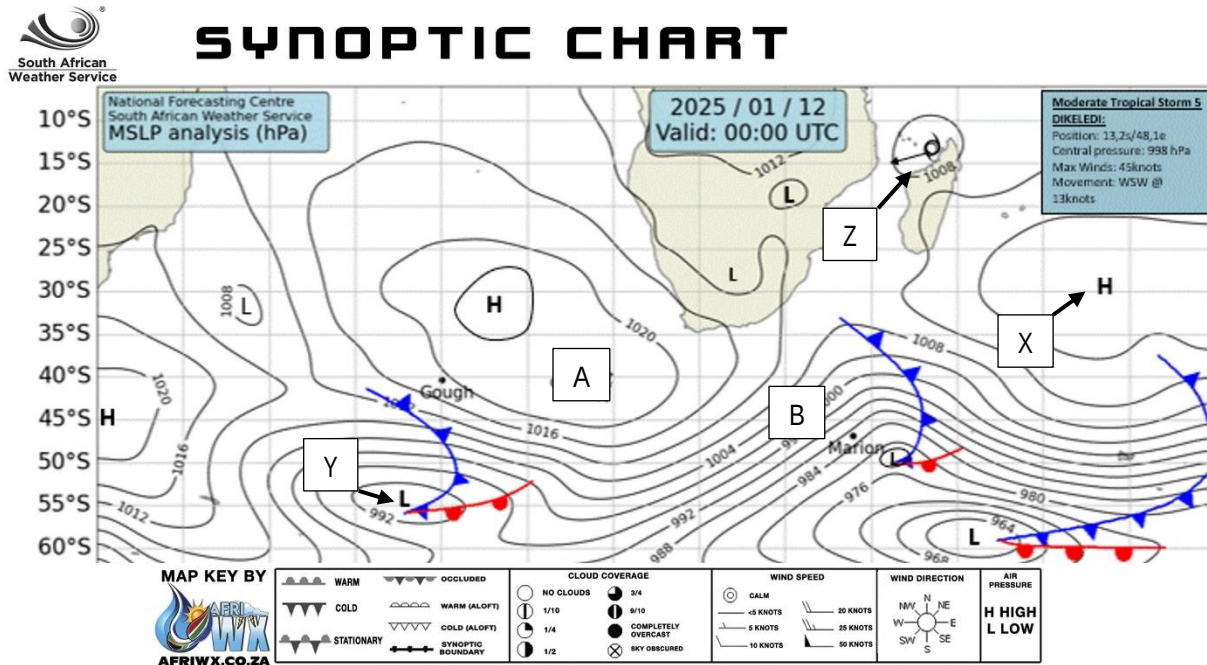
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## **INSTRUCTIONS AND INFORMATION**

1. The question paper consists of FOUR questions.
2. ALL diagrams are included in the question paper.
3. Where possible, illustrate your answers with labelled diagrams.
4. Leave a line between subsections answered.
5. Number your answers correctly according to the numbering system used in this question paper.
6. Write neatly and legibly.

## Question 1 – Short Questions [15]

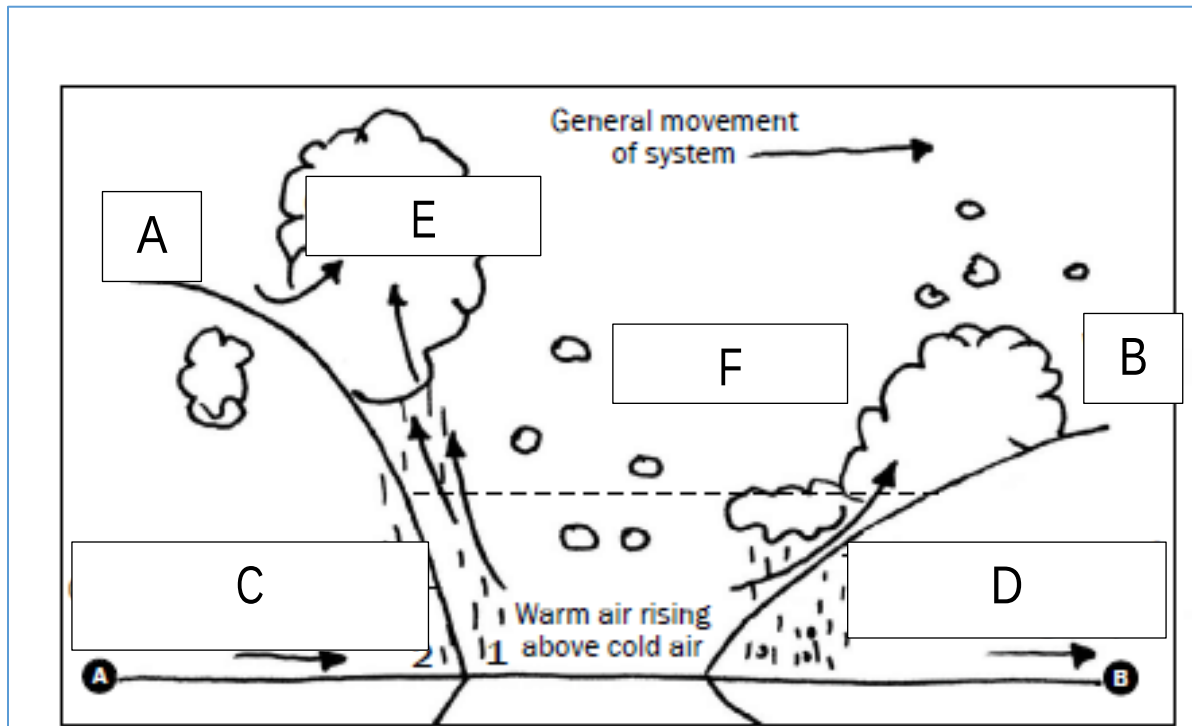
1.1. Refer to the diagram below showing a valley. Answer the questions that follow by choosing only the correct term in brackets. E.g. 1.9. Weather



- 1.1.1. The season shown in the synoptic chart is (Summer/Winter/Autumn/Spring).
- 1.1.2. The high pressure at X is the (South Atlantic/South Indian/Kalahari) high pressure.
- 1.1.3. (Z/Y) is a mid-latitude cyclone.
- 1.1.4. (Z/Y) is a tropical cyclone.
- 1.1.5. The wind speed at B is fast. We know this because it has (a steep pressure gradient/Coriolis force is powerful here/it has both a strong pressure gradient force and a strong Coriolis impact).
- 1.1.6. The conditions at A are likely (clear and calm/wet and windy/cold and humid).
- 1.1.7. Tropical storm Dikeledi is the (2<sup>nd</sup>/3<sup>rd</sup>/4<sup>th</sup>) cyclone of the season.
- 1.1.8. Tropical storm Dikeledi, according to the chart, is currently over (Mozambique/Madagascar/South Africa).

(8x1) (8)

1.2. Refer to the figure below and statements that follow. Select only the correct word in brackets in each statement to make the statement TRUE.

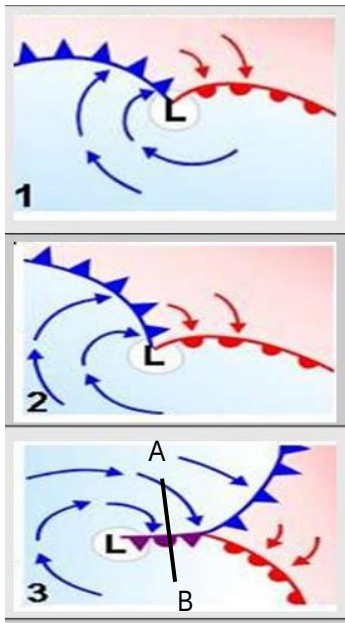


- 1.2.1. The front at B is the (warm / cold / occluded) front.
- 1.2.2. The temperatures of the air found at C, F and D are (cool, cold and warm / cold, warm and cool / warm, cold and cool) respectfully.
- 1.2.3. The symbol for the front found at A is (semi-circles / triangles / a combination of semi-circles and triangles).
- 1.2.4. The clouds that typically form at the front found at A are (cumulonimbus / stratus / cirrus) clouds.
- 1.2.5. The clouds that typically form at the front found at B are (cumulonimbus / nimbostratus / cirrus) clouds.
- 1.2.6. The pressure found at E is (low / high / no) pressure.
- 1.2.7. Once the front labelled A passes over a point, the pressure in that area will (increase / decrease / remain the same).

(7x1) (7)

## **Question 2 – Mid-latitude Cyclones [15]**

Refer to the diagram below and answer the questions that follow:



- 2.1. **Provide** the labels for the stages found at 1, 2 and 3 respectively. (3x1) (3)
- 2.2. **Draw** a fully labelled cross section from A to B in stage 3. This is a cold front occlusion. Make sure you have the following labels:  
 Heading  
 Front symbols  
 Temperature of air pockets  
 Cloud types (4x1) (4)
- 2.3. **Why** are mid-latitude cyclones more common in winter along the southern coast of South Africa? (1x2) (2)
- 2.4. **Discuss TWO** effects that a mid-latitude cyclone would have on farmers in the Western Cape. (2x2) (4)
- 2.5. **Explain** what causes a mid-latitude cyclone to move from the south west towards the east of South Africa. (1x2) (2)

### **Question 3 – Tropical Cyclones [15]**

Refer to the write-up of Tropical Cyclone Dikeledi below and answer the questions that follow:

#### **Disaster description**

On 11 January, tropical cyclone DIKELEDI passed over the far northern Madagascar, on the following day it passed south of Mayotte (France) and Comoros. On 13 January at 6.00 UTC its centre was located offshore over the Mozambique Channel, approximately 75 km east of the coast of the Nampula province, northeastern Mozambique, with maximum sustained winds of 137 km/h. Media report, as of 13 January, three fatalities, and around 15,200 displaced people across northern Madagascar. In addition, the Interdepartmental Crisis Management Operational Centre (COGIC) reports approximately 20,000 evacuated people across Mayotte. DIKELEDI is forecast to pass along the coast of the Nampula province on 13 January, with maximum sustained winds up to 155 km/h. After that, it is expected to continue moving southward over the Mozambique Channel on 14-15 January, further strengthening. Over the next 96 hours, very heavy rainfall, strong winds and storm surges are forecast over most of Madagascar. The same conditions are also forecast over the next 48 hours for Mayotte, Comoros, and the central-northern coast of Mozambique. (ECHO, 13 Jan 2025)

After its passage on 11 January over northern Madagascar, national authorities report that the death toll stands at three people, 352 people have been displaced and 5,803 affected across Analanjirofo, Diana and Sava regions, northern Madagascar.

Initial reports indicate that more than 7,000 people have been affected by Tropical Cyclone Dikeledi in Madagascar. Over 1,000 houses have been damaged, 203 destroyed. Schools resumed on 13 January in some areas in the north of Madagascar. However, school courses for over 8,000 children have been disrupted as nearly 250 classrooms were damaged or destroyed by winds.

- 3.1. **Which** ocean did Cyclone Dikeledi pass through while moving towards Mozambique? (1x1) (1)
- 3.2. **What** was the maximum sustained wind speed recorded as Cyclone Dikeledi approached the Nampula province in Mozambique? (1x1) (1)
- 3.3. **What** will happen to Cyclone Dikeledi after it moves into the southern Indian Ocean? (1x1) (1)

- 3.4. **Define** the term storm surge and explain why it is a significant hazard associated with Cyclone Dikeledi. (2x2) (4)
- 3.5. **Discuss TWO** economic impacts that Dikeledi would have on Madagascar, and **provide TWO** solutions that Madagascar could implement to reduce the impact of future cyclones. (4x2) (8)

#### **Question 4 – Application [15]**

Read the cartoon below and answer the questions that follow:



- 4.1. **What** is 'climate denial' as illustrated in the cartoon? (1x2) (2)
- 4.2. **What** is the name given to these storms when they develop over the Pacific Ocean? (1x1) (1)
- 4.3. **How** many hurricanes, including Milton, have occurred in this hurricane season? (1x2) (2)
- 4.4. **How** does the cartoonist use irony to emphasize the impact of climate change? (2x2) (4)
- 4.5. **What** role do media and politicians play in shaping public opinion about climate change, as suggested by the cartoon? (2x2) (4)
- 4.6. Hurricane Milton became the 5<sup>th</sup> largest hurricane ever recorded on planet Earth. **Explain** how rising sea surface temperatures contribute to the formation of stronger hurricanes like Hurricane Milton. (1x2) (2)

**GRAND TOTAL: 60 MARKS**